Sandeep Seegehalli Akkalappa

Software Engineer

Stuttgart, Germany | +49 176 6036 9964 | sandeepseegehalliakkalappa25@gmail.com
linkedin.com/in/sandeep-seegehalli-akkalappa | sandeepgowda07.github.io

Professional Summary

Software Engineer with 3+ years of experience building production systems from the ground up. I've shipped an AI-powered robotic control platform with real-time voice interaction, deployed cloud-native APIs on AWS, and built automated test pipelines that run on every single commit.

Technical Skills

Programming Python, Java, TypeScript, SQL

Full-Stack FastAPI, React, Spring Boot, REST APIs, WebSockets, Async Programming, MQTT

AI & ML PyTorch, DQN, LLM Integration, Ollama

DevOps Docker, Kubernetes, Git, GitLab, GitHub Actions, AWS, Linux (Bash), CI/CD

Testing Pytest, Unit Testing, Integration Testing, Regression Testing, Test Automation

Work Experience

01/2025 – 01/2026 **Software Developer (Working Student)**, Fraunhofer IPA, Stuttgart

- Built an AI-powered robotic control platform with chat/voice interaction for a pick-and-place robot using React and WebSockets, achieving <3s response latency.
- Integrated LLM-based intent parsing to convert natural language commands into 3 structured robot actions, reducing manual command scripting effort by ~60%.
- Implemented async WebSocket communication between frontend and backend workers by offloading blocking bus calls to a thread pool to keep the event loop free.
- Shipped a production-ready AI demonstrator as a standalone executable, reducing setup time to under 5 minutes.
- Enhanced the CARA robot safety platform with customer-requested features and resolved production bugs, improving software stability.
- Automated the test pipeline using GitLab CI and pytest, ensuring every commit is validated before merge and saving ~3 hours/week of manual QA.
- Developed a PyQt-based demo UI guiding industry visitors through the full CARA safety workflow at Automatica Messe 2025 (Europe's largest robotics fair).

10/2024 – 03/2025 **Research Assistant (Working Student)**, University of Stuttgart, Stuttgart

- Developed a ROS node automating object feeding onto a conveyor belt, synchronizing motion with robot pick cycles to prevent mid-pick collisions.
- Designed control logic coordinating sensor signals, conveyor movement, and robot arm operations for uninterrupted pick-and-place throughput.
- Supervised and evaluated Ada automation experiments for 20+ students across 6 performance criteria.

07/2023 – 07/2024 **Software Engineer (Working Student)**, BOSCH, Feuerbach

- Authored 50+ automated unit tests for Linux CLI tools processing point-cloud and CSV files, improving validation coverage by ~35% across 3+ input formats.
- Containerized the test environment using Docker, ensuring 100% reproducible execution and eliminating environment-specific failures.

12/2020 – 09/2022 **Analyst - Production Support**, Birlasoft, Bangalore, India

- Ensured 99.9% system uptime by monitoring enterprise workflows using Control-M and AWS infrastructure.
- Diagnosed and resolved 30+ production incidents via log analysis and Jira, maintaining average resolution time under 1 hour and minimising SLA breaches.

Projects

03/2026 – 04/2026 **Cloud-Native API Deployment on Kubernetes**

- Deployed FastAPI application on AWS EKS using Helm and Terraform, enabling scalable and repeatable deployments.
- Automated the full delivery lifecycle via GitHub Actions CI/CD and enforced security gates using Trivy to detect and block critical CVEs pre-deployment.

07/2025 – 01/2026 **Master Thesis: Dynamic Function Offloading in Connected Vehicles**

- Designed a slack-aware DQN framework in PyTorch for just-in-time edge/cloud server selection, ensuring deadline compliance in autonomous vehicle workloads.
- Outperformed PSO in scalability where DQN maintained near-constant decision time across 2 to 15 servers while PSO grew proportionally with server count.
- Validated on a real pipeline with actual edge servers and object detection workload, with all tasks completing within deadline across all test scenarios.

07/2024 – 01/2025 **Research Project: Energy Profiling in Service-Oriented Robotics**

- Implemented a ROS-based energy profiling service to measure CPU/GPU utilization, memory usage, and power consumption.
- Orchestrated multiple ROS nodes using tmux scripting for automated launch and debugging
- Integrated MQTT and Home Assistant dashboard for real-time monitoring of energy metrics in service-oriented robotics.

02/2021 – 04/2021 **Order Management System**

- Built a full-stack web app with role-based access for Admin (inventory) and User (ordering) with secure authentication.
- Containerized the app with Docker and deployed on Render, connected to a remote Aiven MySQL cloud database.

Education

10/2022 – 03/2026 **M.Sc. Information Technology**, *University of Stuttgart*, Germany, **Grade: 1.5**

Languages

English (C1) | German (A2) | Kannada (Native)